Installation and use of LaserbeamShapingFoam

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Introduction and Installation

An open-source solver laserbeamFoam [1] was extended to model beam shaping of laser powder bed fusion. The new solver is named as LaserBeamShapingFoam, which can be downloaded from the link: https://github.com/Baoshun-Zhou/LaserBeamShapingFoam.git. The EUProject-GlobalAM branch is used for beam shaping cases.

Before installing the LaserBeamShapingFoam, it is required to download and install the OpenFOAM-10. After downloading the LaserBeamShapingFoam form the source link above, you will see Folders 'laserHeatSource', 'incompressibleInterPhaseTransportModel', 'VOFTurbulenceDamping', 'Make' and 'Example', and source files. Before compiling codes in the parent folder LaserBeamShapingFoam, you need to compile codes in folder ' VOFTurbulenceDamping' and 'incompressibleInte' in advance. Compiling is implemented via type 'wmake' in the terminal within the corresponding folder.

User manual

Then, users can use LaserBeamShapingFoam to run a model. To learn the workflow, users can copy the folder 'Example' to your own working address. Two preprocess steps include grids generating and filed definition before running. To set properties of grid generation, users can define in the file 'blockMesh' within the folder 'Example\system'. Then grids are generated via typing 'blockmesh' in the terminal within the folder 'Example'. To define the field, users need to define a new folder called '0' and then copy the content of the folder 'initial' to '0'. Fields are defined via typing 'setFields' in the terminal within the folder 'Example'. Note that all kinds of boundary conditions are defined in the folder 'initial'. Users can make changes according to the specific requirements before copying 'initial' to '0'. Final step is to run simulation via type 'LaserBeamShapingFoam' in the terminal of Example. More details will be described in the following.

Files and folders within 'Example'

To explain details for users, we illustrate the structure of the files and folders within folder 'Example', as shown in Fig. 1. Three main folders 'system', 'initial' and 'constant' have been

already prepared in advance. Within each of folders, there are several files which setup kinds of conditions.

In Folder ‘system’, file ‘setFieldsDict’ is used to define the region of gas and material phases within the computational domain; ‘fvSolution’ defines the conditions relevant to solvers such as the type of solver, computational tolerance and smooth type; ‘fvSchemes’ defines the choice of numerical schemes (e.g., Gauss linear, and upwind) for various terms of governing equations; ‘decomposeParDict’ defines the setups relevant to the parallel computation, including the choice of partition method, and the number of subdomains; ‘controlDict’ defines the setups of computation strategy, such as timestep, total computational time duration, CFL number; ‘blockMeshDict’ is able to define the meshing strategies (e.g., uniform, stretching, or dynamic mesh cells), and meshing density.

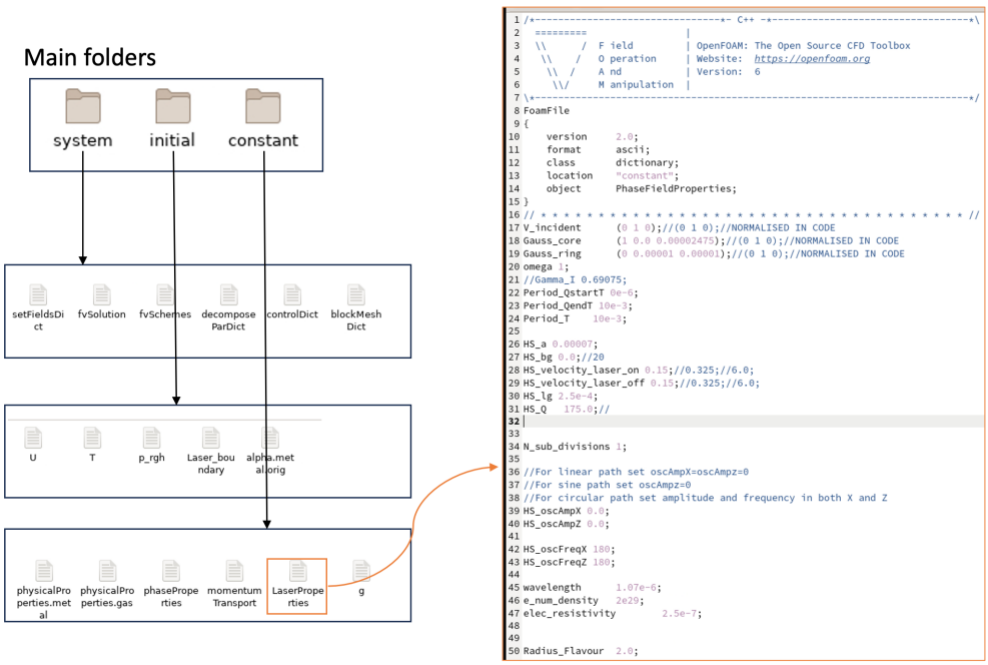


Fig. 1 Folders and files within ‘Example’.

In Folder ‘initial’, we can define various kinds of boundary and internal-field conditions at the initial time, including the fields of velocity, temperature, pressure, laser, and material volume fraction. During the process of simulation, this folder does not participant in computation

directly. After copying this folder to a new folder named as '0' and then typing 'setFields' in the terminal, '0' would participant in the computation.

In Folder 'constant', 'physicalProperties.metal' defines the physical properties for the metal (e.g., thermal conductivity, specific heat, density, solidus and liquidus temperatures); 'physicalProperties.gas' defines the physical properties for the gas phase; 'phaseProperties' defines the interface properties (e.g., surface tension coefficients, Marangoni coefficient, latent heat of vaporation, boiling point); 'momentumTransport' gives the fluid type definition, e.g., laminar;); 'g' defines the gravitational acceleration; and 'LaserProperties' defines the laser properties, and the corresponding explanation is given in the following.

Theory behind and limitations

The solver 'LaserBeamShapingFoam' can model temporal and spatial beam shaping effects of laser, which is implemented by

$$q^{laser} = \mathrm{H}\frac{\eta R^l P}{\pi(a_c + a_r)}\left\{\frac{a_c}{2\sigma_c^2}\exp\left[\frac{R^l(d-\mu_{R1})^2}{2\sigma_c^2}\right] + \frac{a_r}{4\sqrt{2}\omega\mu_{R2}\sigma_r}\exp\left[\frac{R^l(d-\mu_{R2})^2}{2\sigma_r^2}\right]\right\} \quad (1).$$

Referring to Line 17 in Fig. 1, V_incident defines the laser ray direction vector; three components in Line 18 correspond to a_c, μ_{R1} and σ_c in Eq. (1); three components in Line 19 correspond to a_r, μ_{R2} and σ_r in Eq. (1); omega is ω in Eq. (1); In Lines 22-24, these parameters control the laser on and off where $\mathrm{H} = 1$ during time Period_QstartT to Period_QendT and $\mathrm{H} = 0$ during time Period_QendT to Period_T; 'HS_bg' and 'HS_lg' define the initial position of laser; 'HS_bg' is laser power; 'HS_velocity_laser_on' and 'HS_velocity_laser_off' define laser motion speed during laser on and off periods respectively. The parameters in Lines 39-47 define the laser absorptivity based on Frensel's equations, and the details can be seen in Ref. [1]. In Line 50, 'Radius_Flavour' corresponds to R^l in Eq. (1).

So far, the local absorptivity applied in this version of the solver does not change with temperature, and the mass transfer due to evaporation has not been modeled.

References

1. FLINT, T. F., ROBSON, J. D., PARIVENDHAN, G. & CARDIFF, P. laserbeamFoam: Laser ray-tracing and thermally induced state transition simulation toolkit. *SoftwareX* **21**, 101299 (2023).